

Prioritizing Non-Carbon Dioxide Removal Mitigation Strategies Could Reduce the Negative Impacts Associated with Large-Scale Reliance on Negative Emissions

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ABSTRACT: Carbon dioxide removal (CDR) is necessary for reaching net zero emissions, with studies showing potential deployment at multi-GtCO₂ scale by 2050. However, excessive reliance on future CDR entails serious risks, including delayed emissions cuts, lock-in of fossil infrastructure, and threats to sustainability from increased resource competition. This study highlights an alternative pathway—prioritizing near-term non-CDR mitigation and minimizing CDR dependence. We impose a 1 GtCO₂ limit on global novel CDR deployment by 2050, forcing aggressive early emissions reductions compared to 8–22 GtCO₂ in higher CDR scenarios. Our results reveal that this low CDR pathway significantly decreases fossil fuel use, greenhouse gas (GHG) emissions, and air pollutants compared to higher CDR pathways. Driving rapid energy transitions eases pressures on land (including food cropland), water, and fertilizer resources required for energy and negative emissions. However, these sustainability gains come with higher mitigation costs from greater near-term low/zero-carbon technology deployment for decarbonization. Overall, this work provides strong evidence for maximizing non-CDR strategies such as renewables, electrification, carbon neutral/negative fuels, and efficiency now rather than betting on uncertain future CDR scaling. Ambitious near-term mitigation in this decade is essential to prevent lock-in and offer the best chance of successful deep decarbonization. Our constrained CDR scenario offers a robust pathway to achieving net zero emissions with limited sustainability impacts.

KEYWORDS: net zero emissions, carbon dioxide removal, BECCS, DACCS, DORCS, biochar, enhanced rock weathering



1. INTRODUCTION

Global greenhouse gas (GHG) emissions are still rising, reaching record highs in 2022.¹ The world has already warmed by 1.1–1.3 °C above preindustrial levels,^{2–4} and the impacts of climate change are being witnessed.⁵ At the current rates, the remaining 1.5 °C carbon budget will be used up in the 2020s.⁶ Rapid and urgent decarbonization measures are thus needed for the world to stay on track to reach the Paris Agreement climate goals. Near-term decarbonization measures that can quickly be adopted include renewable energy, electrification, carbon-neutral/carbon-negative fuels, energy efficiency, and other demand-side reduction approaches. While these strategies can significantly reduce emissions, they are insufficient on their own to meet climate goals. Residual emissions, typically from hard-to-abate sectors, present a challenge in reaching net zero emissions or net negative emissions. Carbon dioxide removal (CDR) technologies/practices provide an opportunity to offset historical and residual emissions from hard-to-abate sectors.^{2,7–9} In short, CDR is what puts the “net” in “net zero emissions”.¹⁰ Therefore, both CDR and non-CDR mitigation strategies will

be required to limit global mean temperature change to below 1.5 °C by end-of-century.

CDR approaches, especially novel ones, are not yet ready for large-scale deployment because feasibility constraints exist.^{2,11} For existing pathways consistent with 1.5 °C by 2100, large-scale deployment of CDR would be required, reaching tens of gigaton (Gt) scale by 2050.^{2,12–14} Currently, novel CDR capacity is at 0.002 GtCO₂/yr,¹⁵ and this capacity would have to increase by at least a 1500-fold from now to reach around 4.2 (3.7–6.2) GtCO₂ per year by 2050.¹⁶ In perspective, the share of renewables in total energy supply from now to 2050 to achieve net zero emissions would have to increase by 5-fold, while final energy demand from hydrogen increases by 3-fold (between 2030 and 2050).¹⁷ Furthermore, a considerable gap

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exists between policies regarding regional CDR deployment and the estimated multi-Gt scale required by mid-century to deliver the Paris Agreement aspirational goal.¹⁵ Moreover, even if the world had the means of reaching such tens of Gt scale of CDR deployment, it would not be advisable to pursue such pathways for solving climate change when there are other means (less energy–water–land-intensive) of achieving deep decarbonization to limit reliance on CDR. Individually, CDR techniques are characterized by different challenges, and their large-scale deployment should be exercised with caution^{2,18} (Table S1 provides a brief description of the various CDR approaches investigated in this study). Bioenergy with carbon capture and storage (BECCS) and afforestation/reforestation (AR) can be water- and land-intensive and pose threat to food security and biodiversity losses; direct air capture with carbon storage (DACCS) and direct ocean removal with carbon storage (DORCS) are energy-intensive and financially costly; marine CDR approaches such as ocean alkalinity enhancement (OAE) and DORCS could potentially lead to localized acidification, while high concentration of injected CO₂ could be toxic to marine life; enhanced rock weathering (ERW) could potentially increase emissions from water and energy generation.^{2,18–22} Thus, the world must begin implementing several non-CDR mitigation strategies to significantly reduce our reliance on negative emissions and its environmental impacts. Near-term deep decarbonization would ultimately make CDR deployment more effective and feasible in the long term.

While non-CDR mitigation measures like electrification, renewables, hydrogen, and energy efficiency are crucial for reaching net zero emissions, they also come with their own limitations that should not be overlooked. For instance, it is challenging to directly electrify hard-to-abate sectors (responsible for about 30% of global carbon dioxide (CO₂) emissions) by grid and batteries.^{23–25} The intermittency and variability of solar and wind power can challenge grid reliability if there are not adequate storage and flexibility solutions.^{26,27} The significant upfront capital costs of clean energy assets – such as electric vehicles and hydrogen electrolyzers along with renewables integration and transmission expenses also pose affordability issues.^{28,29} To meet global energy and climate goals, the world needs to add or replace 80 million kilometers of power lines by 2040 – this represents the size of the entire global grid built over the past 100 years.³⁰ Hydrogen production currently relies heavily on fossil fuels, generating 2% of CO₂ emissions.³¹ Scaling up hydrogen presents unintended environmental and sustainability issues, as low-carbon hydrogen production requires substantial renewable electricity. This increased electricity demand could compete for limited land and water resources needed for other uses like agriculture.^{32,33} These realities highlight that non-CDR options are also imperfect tools with constraints. Therefore, CDR methods will play an important role in the pathway to net zero by midcentury, despite their drawbacks. As such, there is a trade-off between prioritizing CDR deployment versus non-CDR mitigation in reaching net zero emissions. CDR methods have significant potential to offset residual emissions but face feasibility constraints on large-scale deployment and pose threat to global sustainability.² Non-CDR strategies can achieve deep emissions cuts, but the technologies are currently more expensive than carbon-based options.²⁹

The importance of CDR in reaching global climate milestones should not be undermined. However, excessive

reliance on future CDR deployments should not be used as an excuse to delay phasing out fossil fuels now. An attempt to remove large quantities of CO₂ from the atmosphere will extend the remaining carbon budget,³⁴ delaying the necessary transition away from fossil fuels to achieve climate targets. Existing 1.5 °C-consistent pathways under high CDR especially DACCS are characterized by delayed emission cuts.^{12,14,35} For instance, Fuhrman et al. show that in China's pathway to carbon neutrality, the urgency to transition away from conventional to new energy vehicles reduces under scenarios where DACCS is available to offset emissions.³⁵ By allocating resources, time, and efforts to develop and implement CDR at large scale, there is a missed opportunity to allocate those resources toward other, potentially more effective, zero-carbon technologies or strategies.³⁶ For example, renewables are highly effective at reducing CO₂ emissions and mitigating international conflicts related to energy resources. Additionally, they play a significant role in reducing various negative environmental impacts, including fossil air pollution, mining, and pollution associated with transportation.³⁶ Similar conclusions favor the use of wind and solar power for battery electric vehicles over replacing gasoline vehicles with ethanol–gasoline blends (such as E85), where a portion of the CO₂ produced during ethanol's production is captured and transported.³⁷ Near-term deep decarbonization remains critical, and CDR should complement rather than replace ambitious mitigation policies before mid-century. It is much more straightforward and easier to stop emitting in the first place, compared to trying to remove CO₂ from the atmosphere to achieve climate goals. When deployed responsibly and at scale later this century, CDR can make net zero goals more feasible.³⁸

This study explores scenarios that prioritize non-CDR mitigation over CDR deployment to quantify the implications of this trade-off on the global energy–land–water system. CDR deployment in the current study has been significantly limited by assuming that the world may not reach tens of Gt scale of CDR by 2050, as proposed in previous studies.^{2,12–14,39–42} In the current study, to reach below 1.5 °C by 2100, gross novel CDR deployment of 1 GtCO₂ by 2050 is considered to represent a pathway where the world prioritizes non-CDR mitigation strategies to reach climate targets. Additional scenarios that emphasize CDR approaches, where gross novel CDR scales to between 8 and 22 GtCO₂ by 2050 have been included (see *Methodology*). This study explores the economy-wide transformations required to limit warming to 1.5 °C under varying levels of CDR reliance. With uncertainties around rapidly scaling novel CDR to currently proposed levels by mid-century,^{2,15} we examine mitigation pathways that prioritize deep near-term decarbonization policies over-reliance on future negative emissions. In the absence of multi-Gt negative emissions by 2050, our analysis centers on quantifying the increase in renewable energy, electricity, and hydrogen consumption required across the energy system. We assess the magnitude of emission reductions required from different sectors under such conditions. Mitigation costs and land–water–food–fertilizer implications are also revealed. Additionally, the regional distribution of CDR deployment in our pathways is quantified.

Existing studies highlighting the negative impacts of excessive reliance on CDR technologies have mostly been qualitative and philosophical in nature.^{43–49} An initial attempt to quantitatively reveal the detrimental implications of over-

dependence on CDR for the global energy and climate system has been made by Holz et al.⁵⁰ and Grant et al.^{51,52} Our work builds on these previous efforts and takes an important step further. We expand the quantification of impacts on not just the global energy sector but also its land and water system. The potential trade-offs for food and fertilizer demand, co-benefits such as health risks, and mitigation costs are also revealed in our study. This provides compelling evidence across multiple sectors on the benefits of prioritizing deep decarbonization over excessive CDR deployments. Our findings reinforce the notion that CDR should complement, not replace, aggressive near-term emissions reductions.

Our study explores CDR methods beyond AR and BECCS. Existing integrated assessment modeling (IAM) studies over the past decade have predominantly focused on these two techniques,^{53–57} with some recent attention to DACCS.^{12,14,18,33,35,58} We contribute to this emerging research area by incorporating three additional novel CDR approaches. To the best of our knowledge, only Fuhrman et al.,¹³ Holz et al.,⁵⁰ and Streifer et al.⁵⁹ have gone beyond BECCS–DACCS–AR in terms of IAM research. In this study, we use a modified version of the Global Change Assessment Model (GCAM-TJU), with the ability to deploy six different CDR approaches, i.e., AR, BECCS, DACCS, biochar, ERW, and DORCS, with the last three not included in the public release version of the model.

2. METHODOLOGY

2.1. Model Description. One of the leading IAMs in climate change research is the Global Change Assessment Model (GCAM). The linkages among the world economy, energy system, land use, and environment have been widely investigated using GCAM. “GCAM is a market equilibrium model with a global scope and operates from 1990 to 2100 in 5-year time steps. It can be used to examine, for example, how changes in population, income, or technology cost might alter crop production, energy demand, and water withdrawals, or how changes in one region’s demand for energy affect energy, water, and land in other regions. In all cases, the GCAM core represents the entire world, but it is constructed with different levels of resolution for each of these different systems. In the current release version of GCAM, the energy–economy system operates at 32 regions globally, land is divided into >300 subregions, and water is tracked for 233 basins worldwide. The Earth system module operates at a global scale”.⁶⁰ GCAM has a long list of historical milestones; the tool has been used in virtually every major climate/energy/economics assessment over the last two decades.⁶¹

The objectives of the current study have been achieved using a modified version of GCAM 5.4, GCAM-TJU. Following the development of Fuhrman et al.,¹³ our version of GCAM used in this study has the ability to deploy biochar, ERW, and DORCS in addition to the CDR approaches available in the public release version.

Detailed model description, modeling equations, and parameters are provided in the [Supporting Information](#).

2.2. Scenario Formulation. In our pathways, deep decarbonization is pursued, where net-zero CO₂ emissions are achieved before 2050, consistent with limiting warming to 1.5 °C. To achieve this, two separate markets are developed for Asia and the rest of the world, where both achieve net-zero GHG emissions by 2050, following a peak in emissions before 2025. Thus, in all of our pathways, global net-zero CO₂

Table 1. Scenario Description

scenario	types of CDR	constraint on biomass consumption	LUC emissions subject to constraint	SSP	description
central scenario	-	-	-	-	for a well below 1.5 °C by 2100, global GHG emissions for Asia and the rest of the world peak before 2025 and linearly decline to reach net zero by 2050. Global net zero CO ₂ emission is thus reached before 2045. In perspective, for scenarios assessed in IPCC’s AR6 that limit warming to 1.5 °C with no or limited overshoot (C1a), the 5th–95th percentile for net-zero CO ₂ year is between 2035 and 2070
22 Gt	AR, BECCS, DACCS, DORCS, biochar, ERW	no	yes	2	follows central scenario. Gross novel CDR deployment endogenously increases to 22 GtCO ₂ by 2050
8 Gt	AR, BECCS, DACCS	no	yes	2	follows central scenario. Gross novel CDR deployment is limited to 8 GtCO ₂ by 2050
1 Gt	AR and DACCS (AR and BECCS for sensitivity)	yes	yes (no for sensitivity)	2 (1 and 5 for sensitivity)	follows central scenario. Gross novel CDR deployment is limited to 1 GtCO ₂ by 2050

emissions are attained before 2045. In perspective, in about 50 different pathways assessed in the IPCC's sixth assessment report (AR6), for scenarios that limit warming to 1.5 °C with no or limited overshoot (C1a), the 5th–95th percentile for net-zero CO₂ year is between 2035 and 2070.²

Our key objective in this work is to highlight the potential benefits and trade-offs associated with prioritizing non-CDR mitigation strategies such as renewable energy, electrification, carbon-neutral/negative fuels, and energy efficiency, over excessive reliance on CDR. Three main scenarios are explored in this study, representing different levels of CDR reliance for addressing climate change.

(1) In a highly CDR-dependent pathway, six different CDR approaches are deployed with no limit on how much CO₂ is removed from the atmosphere at any point in time. In this pathway, the gross novel CDR endogenously reaches 22 GtCO₂ by 2050.

(2) In a moderate CDR-dependent pathway, only BECCS and DACCS are deployed (for novel CDR), and their removal potential is limited to 8 GtCO₂ by 2050. We consider BECCS and DACCS due to the significant attention received by these two CDR techniques in existing IAM-based studies compared with the other novel CDR approaches. 8 GtCO₂ by mid-century is consistent with the scenarios assessed in the AR6, while 22 GtCO₂ is on the extreme level of the AR6-assessed CO₂ removals (see refs 2 and 62). In perspective, existing literature project total gross CDR by 2050 at 3.5–30 GtCO₂/yr.^{2,13,14,17,18,35,50,59,62–64}

(3) For a scenario with less emphasis on CDR as a mainstream climate change solution, only DACCS is deployed (for novel CDR), and its total removal potential is limited to 1 GtCO₂ by 2050. DACCS was chosen instead of BECCS due to the latter's high land and water intensity. However, Tables S2–S4 show the implications if BECCS were selected instead of DACCS in this scenario.

It is also worth noting that under very limited availability of negative emissions, bioenergy consumption tends to increase significantly to accelerate deep decarbonization.³⁸ An increase in bioenergy consumption under limited CDR could lead to higher land allocation for cultivating bioenergy crops at the expense of other agricultural lands and water consumption in the process. As such, limiting CDR excessively could indirectly create other sustainability issues regarding food and water security. To mitigate this problem, the pathway with highly limited novel CDR (1 GtCO₂ by 2050) is modeled under constrained bioenergy consumption. In this pathway, bioenergy consumption is constrained to 200 EJ by 2050. It is worth noting that the biomass constraint below 200 EJ by 2050 in this pathway presented modeling feasibility issues. Nonetheless, our consideration of 200 EJ is within the 40–312 EJ range of biomass consumption by 2050 in about 85 1.5 °C-consistent pathways assessed in the AR6.²

In our constrained GHG emission pathways, the constraint on carbon emissions applies to those from fossil fuels and industry; thus, net emissions from land use change are excluded. That is, we do not constrain the net emissions between afforestation and deforestation. However, we priced the carbon emissions from land use at an increasing fraction of the fossil emissions carbon price. This approach not only allows additional emission reductions but also encourages carbon storage through land use change (via afforestation/reforestation).⁶⁵ Tables S5–S7 compare this effect on our

results to a scenario where carbon emissions from land use are not subjected to any form of constraint.

Shared Socioeconomic Pathways (SSPs) are scenarios describing alternative future global developments up to 2100 years without considering climate change or climate policies. They provide narratives and quantifications of socioeconomic factors such as population, urbanization, economic growth, education, and technology. SSPs provide qualitative narratives and quantitative inputs to IAMs to generate integrated scenarios for assessing emissions, climate change, and mitigation pathways. IAMs combine SSP-based socioeconomic assumptions with climate-policy scenarios. This allows them to explore mitigation pathways consistent with temperature targets like 1.5 or 2 °C. Five SSPs exist today in climate change research.^{66–68} Our scenarios are modeled under SSP2 (which is taken as a central 'best-estimate' case,⁶⁹ representing a world where future social, economic, and technological trends do not shift significantly from historical patterns.⁶⁸ In Tables S8–S10, we compare SSP2-based results to those from SSP1 (sustainability) and SSP5 (fossil fuelled development). Here, we are only interested in SSPs characterized by low to high mitigation challenges with moderate to low adaptation challenges. As such, we do not consider SSP3 (regional rivalry) and SSP4 (inequality).^{66–68} Table 1 summarizes the scenarios explored in this study.

3. RESULTS AND DISCUSSION

The implications of varying levels of CDR reliance on the global energy–land–water system are investigated in this

Table 2. Implications of Different Levels of CDR Reliance in Our 1.5 °C-Consistent Pathways on the Global Energy Land Water System^a

indicator	1 Gt	8 Gt	22 Gt
Positive GHG Emissions/Negative CO ₂ emissions			
positive CO ₂ emissions (GtCO ₂)	127.67	127.57	143.67
positive non-CO ₂ emissions (GtCO ₂ e) ^b	61.71	65.38	61.98
CDR by land use change (GtCO ₂)	17.75	16.34	13.36
CDR by novel approaches (GtCO ₂)	3.14	32.36	56.47
Air Pollutants			
CO emissions (Tg)	2958.70	3093.86	3156.44
SO ₂ emissions (Tg)	179.84	194.32	217.76
other air pollutants (Tg) ^c	1841.73	1923.22	1883.63
Primary Energy			
fossil fuel (EJ)	1596.75	2001.73	2417.30
biomass (EJ)	851.57	854.64	858.73
renewables (EJ)	741.91	686.23	608.55
nuclear (EJ)	393.51	325.47	260.22
Final Energy			
electricity and hydrogen (EJ)	1258.29	1209.81	1134.67
carbon-based fuels (EJ)	1450.87	1674.21	1943.35
Land, Water, and Fertilizer			
land allocation for energy and negative emissions (Mkm ²)	197.71	198.80	232.12
cropland allocation (Mkm ²)	62.32	63.25	32.02
water consumption for energy and negative emissions (km ³)	1265.00	1243.69	1279.43
fertilizer demand for energy and negative emissions (MtN)	107.38	110.93	134.60

^aResults are reported in cumulative values from 2025 to 2050 in 5-year interval. ^bInclude; Other GHG (i.e., CH₄, F-Gas, N₂O). ^cNO_x, BC, OC, NMVOC, NH₃.

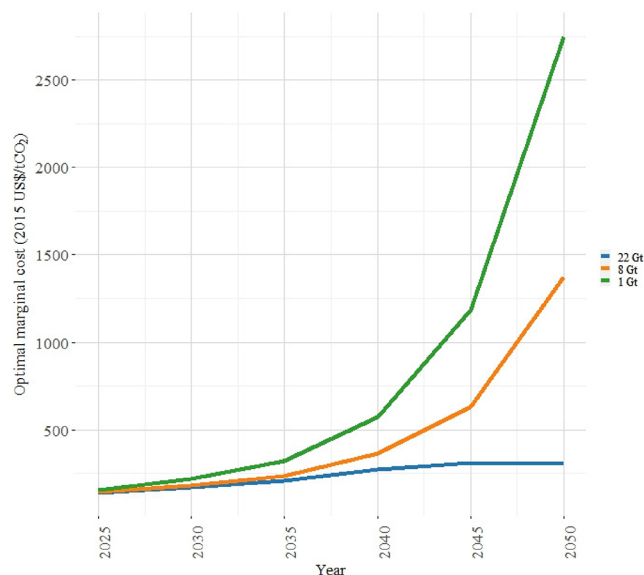


Figure 1. Marginal cost required in each pathway under different novel CDR deployment potentials.

study. Table 2 provides a general summary of the results, highlighting key differences in the implications between the scenarios.

The marginal abatement cost of carbon (also referred to as the carbon price) is the cost of reducing the last unit of emissions to reach a climate target.² If novel CDR can quickly grow to 22 GtCO₂ by 2050, the cost of mitigation would significantly decrease compared to reaching 1 GtCO₂ by 2050. With limited access to CDR deployment, non-CDR mitigation strategies such as clean energy and their technologies would

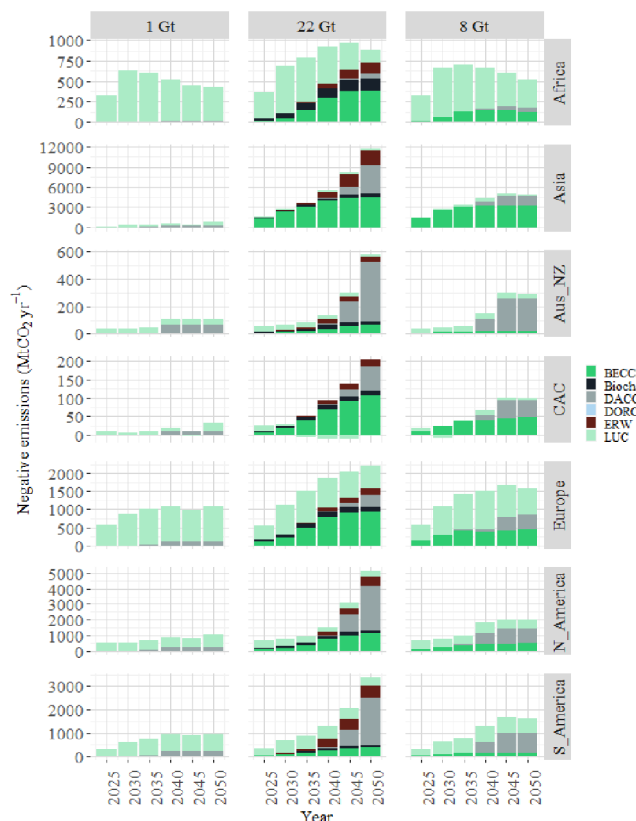


Figure 3. Distribution of CDR deployment by region. Note: CAC – Central America and Caribbean; Aus_NZ – Australia/New Zealand; N_America – North America; S_America – South America.

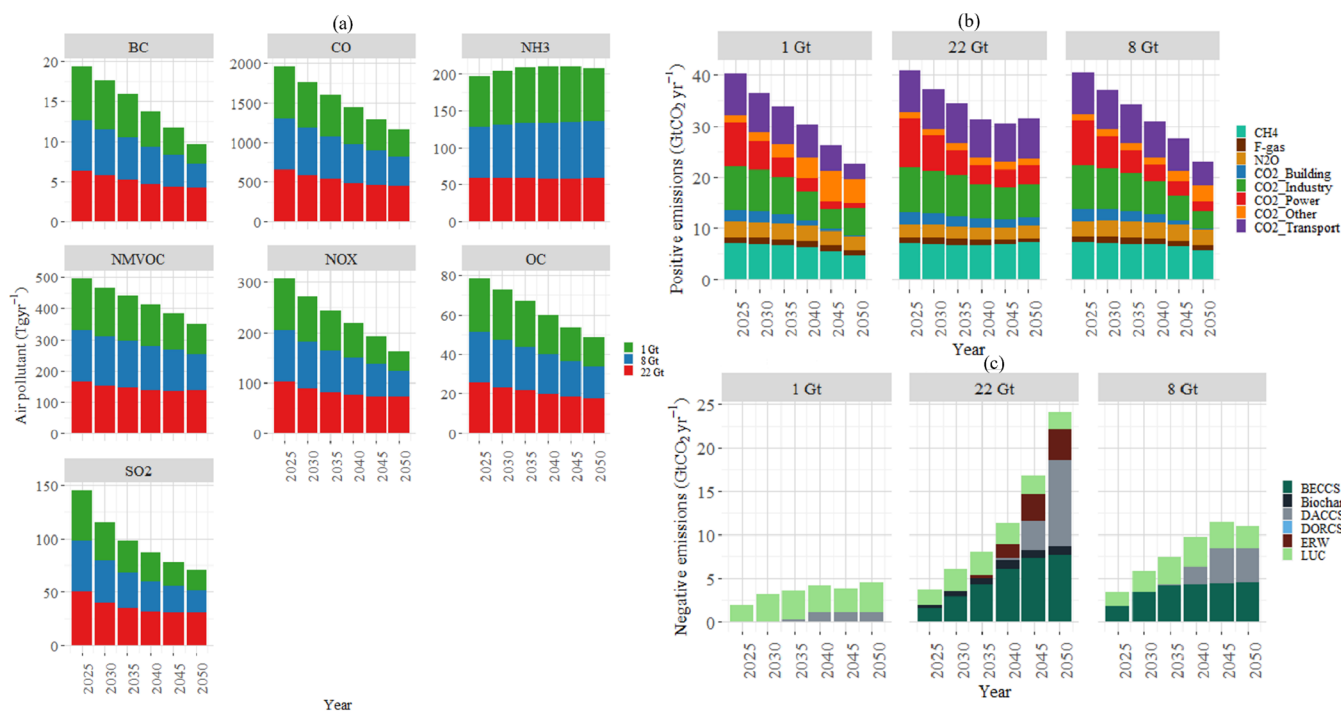


Figure 2. (a) Criteria air pollutants (Tg/yr), (b) positive emissions (GtCO₂ e/yr), and (c) negative emissions (GtCO₂) under different novel CDR deployment potentials. BC: black carbon, NO_x: nitrogen oxides, OC: organic carbon, SO₂: sulfur dioxide, NMVOC: nonmethane volatile organic compounds, LUC: land use change.

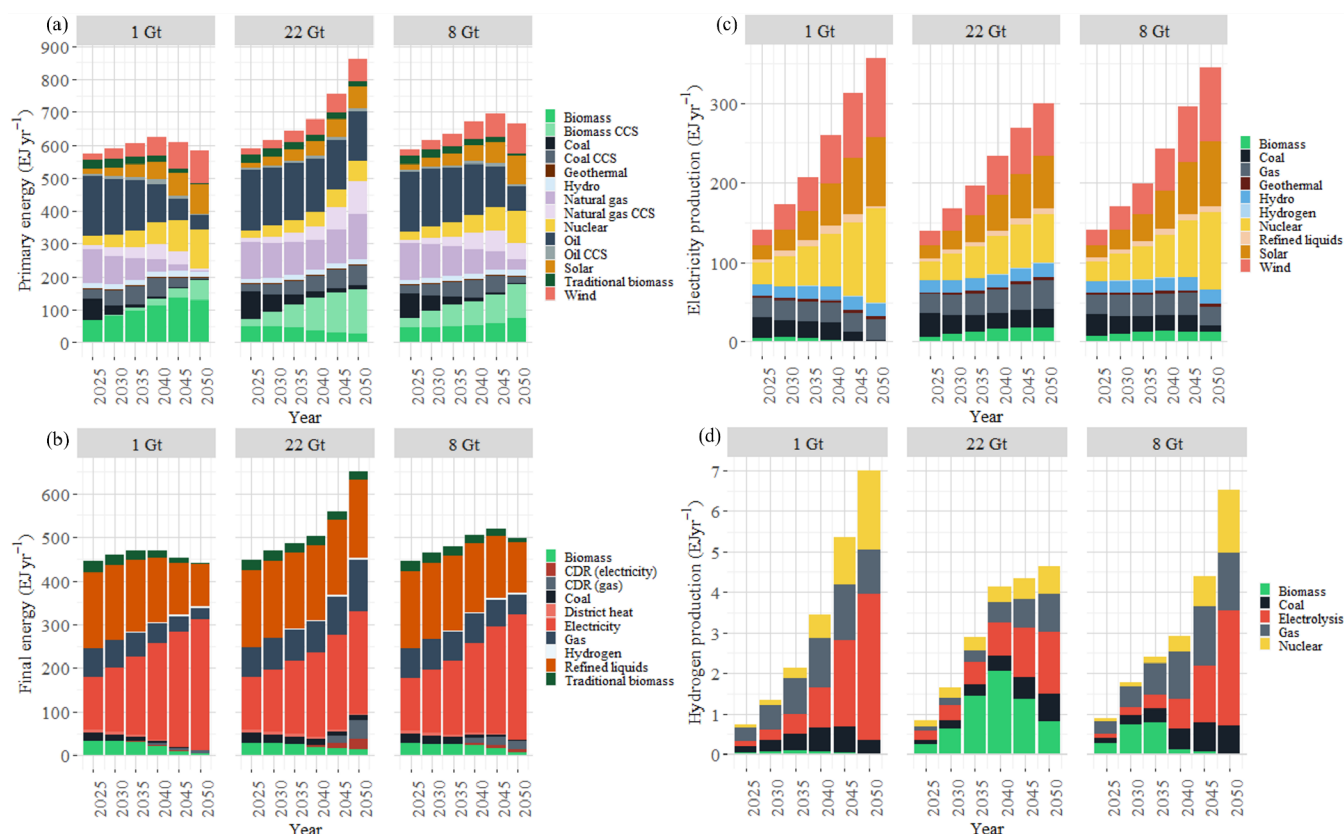


Figure 4. (a) Primary energy consumption by source. (b) Final energy consumption by fuel. (c) Electricity production by fuel. (d) Hydrogen production by fuel under different novel CDR deployment pathways. Refined liquids comprise fossil fuels-to-liquids and biomass-to-liquids (biofuels).

have to be deployed urgently and significantly to reduce emissions and consequently limit the requirement for CDR. However, these clean energy and technologies are relatively expensive options, especially in the near term, compared to their carbon-based counterparts.²⁹ High CDR pathways, on the other hand, present an opportunity to delay the urgent need for rapid decarbonization since emissions can be offset later on to meet the climate target. These reasons explain the expensive mitigation cost for the highly limited CDR scenario, as seen in Figure 1.

If novel CDR grows to 1 GtCO₂ by 2050, the optimal marginal cost required globally to reach our modeled 1.5 °C pathway increases to over \$2700/tCO₂ compared to about \$300/tCO₂ when novel CDR scales to 22 GtCO₂ by 2050. In perspective, AR6-assessed 1.5 °C pathways show a 2050 marginal abatement cost ranging between \$430 and 990/tCO₂.² As decarbonization becomes more difficult and expensive toward the net zero year, the differences in marginal costs across scenarios become more apparent.

Technological breakthroughs are urgently needed to make clean energy and its technologies more cost-competitive, making mitigation via non-CDR approaches economically attractive.

Figure 2 highlights the impact of varying CDR reliance on air pollution, positive and negative emissions under a 1.5 °C pathway. Most air pollutants often result from incomplete combustion processes, such as those in diesel engines and solid fuel burning.^{70,71} Thus, the rapid phase-out of fossil fuels under highly limited CDR deployment (Table 2) would significantly lower the level of air pollutants (Figure 2a). Similarly, with the

increase in clean energy and technologies under very strict CDR deployment, the total GHG emissions are reduced, which could lead to lower warming levels. However, with higher CDR deployment capacities, moral hazard sets in, leading to a slower reduction of GHG emissions and fossil fuel consumption, which could potentially cause the atmosphere to be relatively warmer.

With a priority on non-CDR mitigation strategies, emissions could be reduced significantly (Figure 2b). Between 2025 and 2050, there will be rapid decarbonization especially in the building (~95%), power (~90%), and transport sectors (~60%) by deploying 1 GtCO₂ of novel CDR by 2050, compared to relatively slower decarbonization rates of approximately 30%, 65%, and 5%, respectively, when the world bets on deploying over 20 GtCO₂ of novel CDR by mid-century. Hard-to-abate sectors within the transport and industry sectors would make decarbonization increasingly slower under multi-Gt CDR pathways.^{14,38} It is worth noting that DACCS requires significant amounts of energy, which could increase emissions if supplied by fossil fuels.^{12,14} ERW could potentially increase emissions from water and energy generation. GHG emissions are linked to biochar production, and BECCS comes with significant land requirements and can lead to higher emissions depending on factors such as the scale of deployment, conversion technology, fuel displacement, and the methods and locations of biomass production.² When all of these factors are considered for large-scale CDR deployment, it becomes clear how such pathways could result in higher emissions.

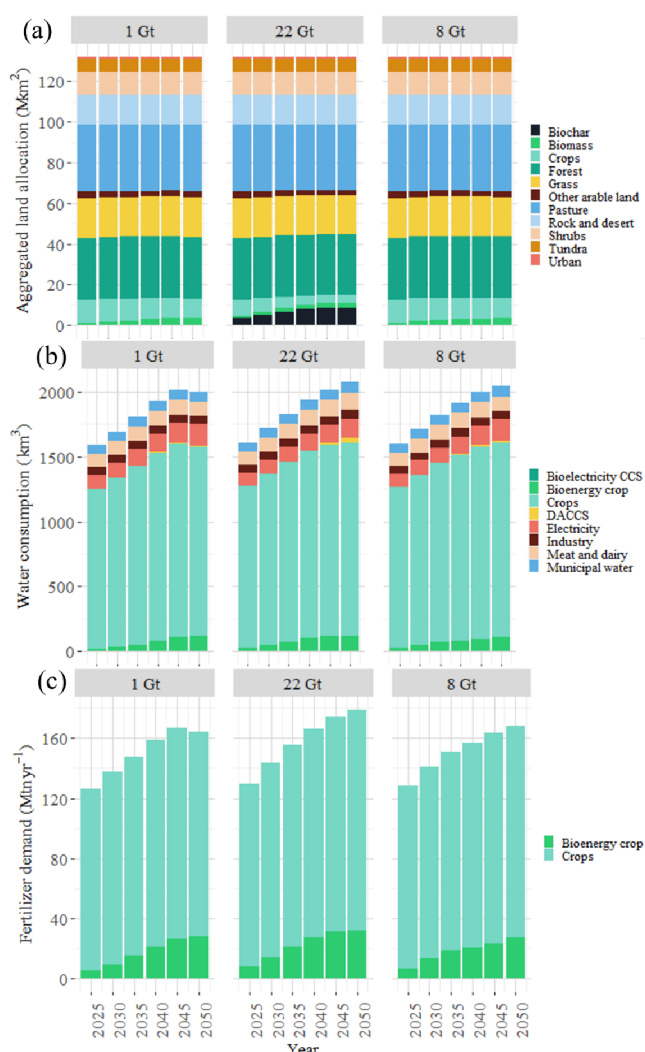


Figure 5. (a) Aggregated land allocation by land use type. (b) Total water consumption by sector. (c) Nitrogen fertilizer demand by crop type under different novel CDR deployment potentials.

Negative emissions by various CDR approaches under the different pathways are reported in Figure 2c. When available, BECCS and DACCS play the most important roles in removing CO₂ from the atmosphere. The role of sequestration by land use change (LUC) is most prominent under highly limited CDR deployment and options. Under the '1 Gt' pathway, net negative emissions from LUC (i.e., the difference between afforestation/reforestation and deforestation) reach 3.5 GtCO₂, which reduces to 1.9 GtCO₂ under the '22 Gt' pathway. When all CDR approaches are available, DORCS will play a very insignificant role due to its high cost.^{13,72,73}

The distribution of CDR by region under each scenario is shown in Figure 3. In all pathways, gross CDR demand would be largely concentrated within Asia, primarily driven by China. While CDR deployment concentrates in Asia across all scenarios based on demand and resource factors, the preferred mix of CDR options varies by region based on comparative costs and availabilities of land, biomass, and other key inputs. BECCS plays the most important role (when all CDR options are available) in Africa, Asia, Europe, and Central America and Caribbean (CAC), but regions like North/South America and Australia would find DACCS deployment relatively cheaper. This could be attributed to constraints around utilizing land for

bioenergy feedstocks needed for BECCS in these regions,^{74,75} which increases costs for land-intensive BECCS, making DACCS more cost-competitive by comparison. In terms of relative competition, biochar deployment in Africa would be more competitive against BECCS and DACCS compared with the observations from other regions. High cropland areas and favorable climatic conditions increase the feasibility of biochar as a CDR option in Africa.

The remaining 1.5 °C carbon budget could be exhausted within the next 5 years at current rates of emissions.¹ This necessitates rapid growth in renewables and nuclear in the global energy supply. Under a highly limited CDR pathway, the combined share of wind and solar in primary energy consumption would be twice as much as that under a multi-Gt CDR pathway (Figure 4a), primarily driven by the slow phase-out of coal consumption (in the high CDR pathway). The slow decarbonization under a high CDR pathway would lead to a continued increase in total energy consumption, while peak consumption occurs before 2045 under low CDR pathways. As coal quickly phases out in the '1 Gt' scenario, electrification of end-use sectors grows rapidly (Figure 4b). In addition, a low deployment of CDR significantly reduces electricity and natural gas consumption for negative emissions. The pathway to well below 1.5 °C under highly limited CDR would see higher renewable and nuclear power generation, reaching over a 90% share in total generation by 2050 (Figure 4c). The International Energy Agency's net-zero pathway shows zero coal electricity generation by 2050.⁷⁶ In comparison, the share of coal electricity generation in our '1 Gt' pathway reaches 0.4% by 2050 compared to nearly 10% in the '22 Gt' pathway. Similar observations are witnessed under hydrogen production (Figure 4d).

The pathway to 1.5 °C would bring about significant changes in land use due to the increase in demand for both energy and negative emissions. By limiting novel CDR deployment to 1 GtCO₂, combined land allocation for bioenergy crop cultivation and forest cover in the form of negative emissions and energy requirements reduces when compared to the high CDR pathway (Table 2), primarily driven by biochar cropland. Under the high CDR pathway, biochar cropland will quickly grow to 8.4 Mkm² by 2050 at the expense of bioenergy crops and food/nonfood crops. Relying heavily on land-based biological CDR approaches to achieve negative emissions could have unintended consequences for food production and resource use. Afforestation, reforestation, biochar, and bioenergy crops would likely displace existing cropland and pastureland.² The '1 Gt' pathway, primarily as a result of the absence of biochar, would record a reduction in food/non food cropland of 20% between now and mid-century compared to 50% in the '22 Gt' pathway (Figure 5a). Interestingly, the absence of biochar and lower net negative emissions from LUC in the '8 Gt' pathway will see food/non food cropland reduction of 15%.

As a result of low CDR demand, the '1 Gt' pathway also leads to a reduction in total water consumption for negative emissions (Figure 5b). Water consumption by DACCS alone reaches below 5 km³ by 2050 under highly limited CDR pathways compared to nearly 40 km³ under a highly CDR-dependent pathway. Due to its high electrification, water consumption for power generation would remain higher in the limited CDR scenario. Overall, when combined, total water consumption for negative emissions and energy generation (for CDR, bioenergy cultivation, primary energy extraction, and

Table 3. Comparison of Results Here with Existing 1.5 °C-Consistent Pathways

indicator	existing literature	current study		
		22 Gt	8 Gt	1 Gt
2050 share of wind, solar, and nuclear in primary energy (%) ^a	0.44–51.52	22.36	41.03	52.47
2050 share of unabated fossil fuel electricity (%)	0.2	2.01	0.01	0
2050 share of wind, solar, and nuclear electricity (%) ^b	1–60	64.33	78.97	85.16
2050 share of electricity in final energy consumption (%)	40–64	39.69	59.25	68.96
2050 GHG emissions (GtCO ₂ e/yr)	0–15	10.40	9.58	8.45
year of net zero CO ₂ emissions	2035–2070	before 2045		
2050 total net CO ₂ emissions (GtCO ₂ /yr)	1.7 to –10.6	–8.89	–8.03	–7.03
change in CH ₄ emissions between 2019 and 2050 (%)	–60 to –45	–23.05	–39.93	–49.44
BECCS deployment by 2050 (GtCO ₂ /yr)	0.4–14.5	7.76	4.50	0
DACCS deployment by 2050 (GtCO ₂ /yr)	0.09–2.2	9.94	3.97	1
AR deployment by 2050 (GtCO ₂ /yr)	up to 10	1.90	2.54	3.51
ERW deployment by 2050 (GtCO ₂ /yr)	0–3	3.56	0	0
land allocation to energy crops by 2050 (Mkm ²)	0.56–4.82	2.38	3.67	3.77
cropland reduction between 2010 and 2050 (Mkm ²)	–2	–7.55	–1.89	–2.40
2050 water consumption for energy and negative emission (km ³)	439.54	305.10	289.78	295.64
cumulative fertilizer demand for bioenergy crop cultivation from 2025 to 2050 (MtN)	190.57	134.60	110.93	107.38

^aNot a combined value for all three energy sources. Wind and solar have a median of 21.13%, a maximum and minimum value of 51.52% and 4.48%, respectively. Nuclear has a median value of 4.49%, a maximum and minimum value of 12.84% and 0.44%, respectively. ^bNot a combined value for all three energy sources. Wind and solar have a median of 19.10%, a maximum and minimum value of 60.11% and 1.65%, respectively. Nuclear has a median value of 8.10%, a maximum and minimum value of 27.53% and 1.02%, respectively.

electricity) would remain lower in the low CDR pathway, as shown in Table 2.

Table 2 shows lower cumulative biomass consumption in the low CDR pathway. In GCAM, biochar is produced from bioenergy crops, crops, and forestry residues.⁷⁷ Considering the high biochar cropland allocation in the ‘22 Gt’ pathway, in addition to its land allocation for bioenergy crop cultivation, the total nitrogen fertilizer demand for growing bioenergy crops remains consistently higher from 2025 to 2050 under the highest CDR pathway, reaching a cumulative demand of 135 MtN compared to 107 MtN in the lowest CDR pathway over the next three decades (Figure 5c).

Table 3 summarizes how our results compare to similar pathways existing in the literature, as a means of validation and benchmarking. The results from existing scenarios can mainly be found in IPCC’s AR6², IEA’s net zero roadmap,⁷⁶ and the work of Fuhrman et al.¹⁸

4. IMPLICATIONS

This study highlights the critical role of prioritizing non-carbon dioxide removal (non-CDR) mitigation strategies like renewables, electrification, carbon neutral/negative fuels such as hydrogen, and energy efficiency in reaching net zero emissions and limiting global temperature increase to well below 1.5 °C. With novel CDR technologies still in their infancy, rapidly scaling their deployment to a multi-Gt scale within the next three decades is highly uncertain.² As such, pathways overly reliant on future CDR deployment could be risky and lead to complacency in near-term mitigation efforts.

By limiting novel CDR deployment to just 1 GtCO₂ by 2050, compared to 8–22 GtCO₂ in higher CDR pathways, we find that a low CDR approach drives rapid phase-out of unabated fossil fuels. Coal and natural gas use will plunge dramatically, with fossil fuel and industry CO₂ emissions reducing by 60% to over 90% within the next three decades. The public health co-benefits due to significantly reduced air pollutants are immense, potentially reducing the risk of lung cancer, stroke, and cardiovascular and respiratory disease

linked to fossil fuel combustion. Sustainability issues related to land use, water consumption, and fertilizer demand also show reduced impacts from the requirement of these resources for energy and negative emissions under limited CDR pathways. Multi-Gt scale deployment of CDR, especially with biochar, reduces food/non-food cropland significantly. Our ‘22 Gt’ pathway shows cropland reduction of over 50% within the next three decades compared to 20% in the ‘1 Gt’ pathway.

However, highly limited CDR pathways come at the cost of higher mitigation expenditures. With limited CDR availability, clean energy technologies must become cost-competitive faster to enable affordable deep decarbonization. Trillions of dollars in upfront capital are needed this decade to scale renewables, batteries, electric vehicles (EVs), hydrogen, grids, and other low-carbon solutions before they become cost-competitive with fossil fuels.⁷⁶ Poorer regions may struggle to finance this transition, risking higher energy poverty if the burden of accessing clean energy falls heavily on consumers.

Going forward, regional-based assessments are needed to reveal potential CDR implications for individual countries or regions within key continents such as Asia, where a large share of future negative emissions would be concentrated.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.est.3c06866>.

Additional modeling details, materials and methods, and results; enhanced weathering supply curves for GCAM regions (Figures S1 and S2); description of CDR approaches (Table S1); energy-based, emission/sequestration-based, and land/water/fertilizer results comparison between BECCS and DACCS scenarios (Tables S2–S4), between constrained and unconstrained land use emission scenarios (Tables S5–S7), between SSP1, 2, and 5 (Tables S8–S10); fraction of CO₂ captured by transformation technologies (Table S11); primary energy transformation technologies default cost assump-

tions (Table S12); electricity technology capacity factors (Table S13); electricity technology capture fractions (Table S14); electricity technology retirement parameters (Table S15); industrial energy use default efficiencies (Table S16); carbon storage resource supply curve points (Table S17); calibration values for the CO₂ removal sector (Table S18); CO₂ capture rates for DAC and process heat DAC technology (Table S19); parametrizations for DACCS technologies (Table S20); enhanced weathering cost adder (Table S21); electrical energy inputs for enhanced weathering (Table S22); electrical energy input derivation for DORCS paired with reverse osmosis desalination (Table S23); non-fuel costs of DORCS (Table S24); non-energy cost assumptions for DORCS paired with reverse osmosis desalination (Table S25); GCAM assumptions for direct ocean capture (Table S26) (PDF)

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Notes

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